KING KHALID UNIVERSITY
Dept. of Computer Science

كلية علوم الحاسب
College of Computer Science

Subject Code: 1312CCS-3
Subject Name: Introduction to Programming
Section No:4377

Proposed Project 1

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Project Grading Criteria (Total: 5 Marks)

Criteria	Description	Weight (%)
1. Problem Understanding & Plan	Clear problem, correct input/output, logical steps (algorithm)	**25%**
2. Python Implementation	Program works correctly, uses input, if-else, correct logic	**35%**
3. Code Quality & Readability	Clean, readable, well-structured code	**15%**
4. Demo (Screenshot / Output)	Clear evidence program runs correctly	**5%**
5. Teamwork & Task Distribution	Fair contribution and clear roles	**5%**
6. Oral Presentation & Understanding	Explanation, logic, and ability to modify code	**15%**

<u>Project Task</u>

Weather App

About the Project:

A Python program that fetches and displays real-time weather data for a user-specified city using the OpenWeatherMap API.

Part 1:

Problem Statement:

The objective of this program is to fetch and display real-time weather conditions for any given city using an API. It runs in a continuous loop, allowing the user to make multiple weather inquiries until they choose to exit.

Inputs and Outputs:

Input: The user will enter a city name (String). They can also type "exit" to quit the program.

Output: The program will display the current temperature (in °C), humidity percentage, and a brief weather description. If the entered city is invalid, it outputs an error message asking the user to try again.

Algorithm (Steps of Solution):

1. Start the program and display a welcome message.
2. Enter a continuous loop (while True) to accept multiple user requests.
3. Prompt the user to input a city name.
4. Check the input: If the user types "exit", print a goodbye message and terminate the loop to end the program.
5. If a city name is provided, construct the API URL using the city name and the OpenWeatherMap API key.
6. Send an HTTP GET request to the API to fetch the weather data.
7. Check the response status:
    - If successful (Status 200): Extract the temperature, humidity, and weather description from the received JSON data. Display these details clearly to the user.
    - If unsuccessful (City not found): Display an error message ("Sorry, city not found ,try again").
8. Repeat the process from step 3.

Part 2:

Screenshots of Output:

```
=== Welcome to Weather App (type 'exit' to quit) ===

Enter the city name: al hasa
==== Weather details ====
City:al hasa
temperature: 19.83 °C
humidity: 46%
description: broken clouds
========================

Enter the city name: exit
Goodbye! :(
```

Python Code:

```python
import requests

API_key ="YOUR_API_KEY_HERE"
BASE_URL = "https://api.openweathermap.org/data/2.5/weather"

print("=== Welcome to Weather App (type 'exit' to quit) ===")

while True:
    city= input("\nEnter the city name: ")

    if city.strip().lower()=="exit":
        print("Goodbye! :(")
        break


    url=f"{BASE_URL}?q={city}&appid={API_key}&lang=en&units=metric"
    response = requests.get(url)


    if response.status_code==200:
        data=response.json()


        temp= data["main"]["temp"]
        humidity=data["main"]["humidity"]
        description=data["weather"][0]["description"]

        print("==== Weather details ====")
        print(f"City:{city}")
        print(f"temperature: {temp} °C")
        print(f"humidity: {humidity}%")
        print(f"description: {description}")
        print("========================")

    else:
        print("Sorry, city not found!,try again")
```